TUTORIAL 5

Linear Dependence, Bases in F^n, and Matrix Operations

5.1 Dependence and Bases in F^m

DEFINITION 5.1

The list $v_1, \ldots, v_k \in F^m$ is linearly independent if

$$\alpha_1 v_1 + \cdots + \alpha_k v_k = 0 \quad \Longrightarrow \quad \alpha_1 = \cdots = \alpha_k = 0.$$

It is dependent if there is a relation with coefficients not all zero. A basis of F^m is an independent spanning list. Let e_j be the column with entry 1 in position j and zero elsewhere. The list $(e_1, \ldots, e_m)$ is the standard basis: $(x_1, \ldots, x_m)^T = \sum_{j=1}^{m} x_j e_j$, and each coefficient is determined by its corresponding component.

PROPOSITION 5.1

A list is a basis if and only if every vector has a unique representation as a linear combination of its entries. Spanning gives existence. Independence gives uniqueness by subtracting two representations. Conversely, uniqueness of the representation of zero implies independence.

NOTATION AND CONVENTIONS

For a collection of columns or matrices closed under addition and scalar multiplication, a basis is an independent spanning list. Its dimension, denoted dim, is the number of elements in a basis. In particular, $\dim F^m = m$ and $\dim M_{m\times n}(F) = mn$. The abstract vector-space definition appears in Tutorial 8.

NOTATION AND CONVENTIONS

Let F be a field. The set $M_{m\times n}(F)$ consists of matrices with m rows, n columns, and entries in F; $M_n(F)$ consists of square matrices of order n. The entry A_{ij} is in row i and column j. The column $e_j \in F^n$ has entry 1 in position j and zero elsewhere. For a square matrix, the trace is $\operatorname{tr} A = \sum_{i=1}^{n} A_{ii}$. The identity matrix I_n has diagonal entries 1 and all other entries 0. The zero matrix has only zero entries.

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5.2 Linear Dependence and Independence

EXAMPLE 5.1

The set

$$\left\{ \begin{pmatrix} 0 \\ -4 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$$

is linearly dependent, since

$$1 \cdot \begin{pmatrix} 0 \\ -4 \\ 1 \end{pmatrix} - 1 \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + 4 \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 0.$$

Rearranging this equation expresses $\begin{pmatrix} 0 \\ -4 \\ 1 \end{pmatrix}$ in terms of the other vectors.

How Do We Test Dependence?

MOTIVATION AND GUIDING IDEA

- Write a **homogeneous** system whose unknowns are the coefficients in the linear combination.
- Form its matrix and row-reduce.
- **Linear independence:** the only solution is the trivial, or zero, solution.
- **Linear dependence:** there is a nonzero solution. Over an infinite field there are then infinitely many solutions.
- A homogeneous system always has a solution: row reduction cannot produce an inconsistent row.

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EXERCISE 5.1

Let

$$v_1 = (1, 1, 0)^T, \qquad v_2 = (0, 2, 1)^T, \qquad v_3 = (3, -1, -2)^T.$$

Please determine linear dependence, please find all dependence relations, and please express one vector using the others.

WORKED SOLUTION

Write $av_1 + bv_2 + cv_3 = 0$. Its coordinates give

$$a + 3c = 0, \qquad a + 2b - c = 0, \qquad b - 2c = 0.$$

Thus $a = -3c$, $b = 2c$, and the middle equation is $-3c + 4c - c = 0$. All relations are

$$(a, b, c) = t(-3, 2, 1), \qquad t \in \mathbb{R}.$$

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EXERCISE 5.1 CONTINUED

In particular, $-3v_1 + 2v_2 + v_3 = 0$, so $v_3 = 3v_1 - 2v_2$. This is a nontrivial dependence. In $av_1 + bv_2 = 0$, the first and third coordinates force $a = b = 0$, so the first two vectors are independent.

We've got it. Victory!

5.2.1 Useful Criteria for Linear Dependence

1. Every set containing the zero vector is linearly dependent.
2. A singleton consisting of a nonzero vector is linearly independent.
3. **Testing a set for linear dependence.**
   - As in the previous tutorial, set a linear combination equal to zero, obtain a homogeneous system, row-reduce, and determine its solutions.
   - Recall: only the zero solution means independence; a nonzero solution means dependence.
   - Another method from the lecture is to place the vectors as the **rows** of a matrix and row-reduce. If a zero row occurs, the original rows are dependent; if there are no zero rows, they are independent.

5.2.2 Spanning Sets

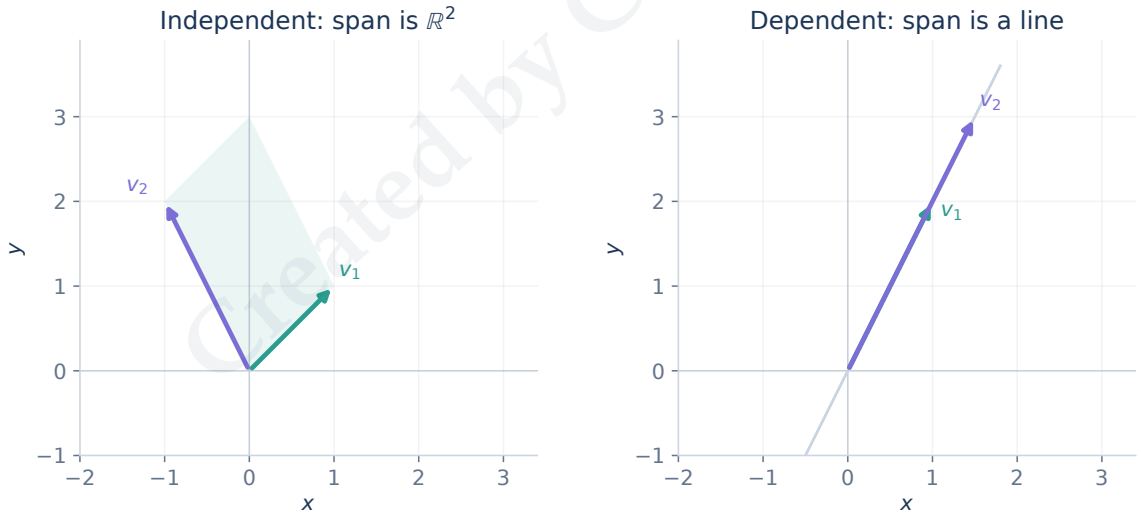


Figure 5.1. Geometric span: a nonzero vector spans a line through the origin; two independent vectors span a plane through the origin.

EXERCISE 5.2 FOUNDATION

Work over $\mathbb{R}$. Let $u = (1, 1, 0)^T$ and $v = (0, 1, 1)^T$.

1. Please describe $U = \text{span}\{u, v\}$ by an equation.
2. Please determine whether $(3, 7, 4)^T$ and $(3, 6, 4)^T$ belong to U, finding coefficients when possible.
3. Please determine whether u, v are independent.

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EXERCISE 5.2 FOUNDATION CONTINUED

WORKED SOLUTION

For $a, b \in \mathbb{R}$,

$$au + bv = (a, a + b, b)^T.$$

Every vector in the span therefore satisfies $y = x+z$. Conversely, if $(x, y, z)^T$ satisfies $y = x+z$, then

$$(x, y, z)^T = xu + zv.$$

Thus $U = \{(x, y, z)^T : y = x + z\}$. For the first vector, $7 = 3 + 4$, and the vector is $3u + 4v$. For the second, $6 \neq 3 + 4$, so it is not in the span. Finally, if $au + bv = 0$, the first and third coordinates give $a = b = 0$. Thus u, v are independent; in particular, the coefficients found above are unique.

That does it. Victory!

EXERCISE 5.3

Please determine whether these vectors form a basis of $\mathbb{C}^3$. If not, please give a dependence relation.

$$v_1 = (1, 0, 2i)^T, \quad v_2 = (1 - i, 1, 0)^T, \quad v_3 = (5 + 2i, 1 + 2i, -2 + 4i)^T.$$

WORKED SOLUTION

The first two vectors are independent: in $av_1 + bv_2 = 0$, the second coordinate gives $b = 0$, and then the first gives $a = 0$. Computing coordinates gives

$$(2 + i)v_1 + (1 + 2i)v_2 = v_3.$$

$$(2 + i) + (1 + 2i)(1 - i) = 5 + 2i, \qquad (2 + i)2i = -2 + 4i.$$

Thus $(2 + i)v_1 + (1 + 2i)v_2 - v_3 = 0$ is nontrivial. The rank is 2, so the triple is not a basis of $\mathbb{C}^3$.

All wrapped up. Victory!

EXERCISE 5.4

Please determine whether these vectors form a basis of $\mathbb{C}^3$. If not, please give a dependence relation.

$$v_1 = (1, 0, i)^T, \quad v_2 = (2 - i, 1, 0)^T, \quad v_3 = (4 - 3i, 2 - i, -1 + i)^T.$$

WORKED SOLUTION

The first two vectors are independent: in $av_1 + bv_2 = 0$, the second coordinate gives $b = 0$, and then the first gives $a = 0$. Computing coordinates gives

$$(1 + i)v_1 + (2 - i)v_2 = v_3.$$

$$(1 + i) + (2 - i)^2 = 4 - 3i, \qquad (1 + i)i = -1 + i.$$

Thus $(1+i)v_1 + (2-i)v_2 - v_3 = 0$ is nontrivial. The rank is 2, so the triple is not a basis of $\mathbb{C}^3$.

We've reached the finish. Victory!

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EXERCISE 5.5

Let $\{u, v, w\}$ be a basis of $\mathbb{R}^3$. Please determine which sets are also bases.

1. $\{u, u+2v, u-2v\}$.
2. $\{u/3, v/5, w/7\}$.
3. $\{u+v+w, u+3v, 2v-w\}$.

WORKED SOLUTION

Each set has three vectors in a space of dimension 3, so independence suffices.

1. No: $-2u + (u+2v) + (u-2v) = 0$ is a nontrivial dependence.
2. Yes. If $a(u/3)+b(v/5)+c(w/7) = 0$, independence of u, v, w gives $a/3 = b/5 = c/7 = 0$, hence $a = b = c = 0$.
3. No: $(u+v+w) - (u+3v) + (2v-w) = 0$ is a nontrivial dependence.

Yes! Victory!

5.3 Matrix Arithmetic

Throughout this section, let

$$A = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 2 \\ -1 & 5 \end{pmatrix}.$$

5.3.1 Scalar Multiplication

Scalar multiplication is defined for every matrix. Multiply **every** entry by the scalar: $(\lambda A)_{ij} = \lambda (A)_{ij}$.

EXAMPLE 5.2

Compute

$$-3A = \begin{pmatrix} -3\cdot 2 & (-3)(-1) \\ -3\cdot 3 & -3\cdot 4 \end{pmatrix} = \begin{pmatrix} -6 & 3 \\ -9 & -12 \end{pmatrix}.$$

5.3.2 Matrix Addition and Subtraction

These are defined only for two matrices of **exactly the same shape**. Add or subtract corresponding entries: $(A \pm B)_{ij} = (A)_{ij} \pm (B)_{ij}$.

EXAMPLE 5.3

Compute

$$A + B = \begin{pmatrix} 2+0 & -1+2 \\ 3-1 & 4+5 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 2 & 9 \end{pmatrix}.$$

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5.3.3 Matrix Multiplication

A product is defined only when the number of columns of the first (left) matrix equals the number of rows of the second (right) matrix. If A is $k \times m$ and B is $m \times n$, then AB is $k \times n$ and is defined by

$$(AB)_{ij} = \sum_{r=1}^{m} a_{ir} b_{rj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{im}b_{mj}.$$

This is the entry in row i, column j of the product. Follow row i of A and column j of B, each containing m entries: multiply matching entries and add. **Note:** the existence of AB does **not** ensure the existence of BA. With the dimensions above, the latter also requires $n = k$.

EXERCISE 5.6

Please compute AB and BA for the matrices defined above.

WORKED SOLUTION

$$AB = \begin{pmatrix} 2\cdot 0 + (-1)(-1) & 2\cdot 2 + (-1)5 \\ 3\cdot 0 + 4(-1) & 3\cdot 2 + 4\cdot 5 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -4 & 26 \end{pmatrix},$$

$$BA = \begin{pmatrix} 0\cdot 2 + 2\cdot 3 & 0(-1) + 2\cdot 4 \\ (-1)2 + 5\cdot 3 & (-1)(-1) + 5\cdot 4 \end{pmatrix} = \begin{pmatrix} 6 & 8 \\ 13 & 21 \end{pmatrix}.$$

Each entry is the sum of the matching row-column products. Reversing the order changes the result.

V-I-C-T-O-R-Y!

NOTE

Important conclusion: matrix multiplication is not commutative. The equality $AB = BA$ need not hold. In the example, $AB \neq BA$ even though both products are defined.

EXERCISE 5.7

Let A be a $k \times m$ matrix with a zero row, and let B be $m \times n$. Please prove that the $k \times n$ matrix AB has a zero row.

WORKED SOLUTION

Let i be the index of a zero row of A, where $1 \leq i \leq k$. Thus $(A)_{ir} = 0$ for every $1 \leq r \leq m$. In row i of AB, for every $1 \leq j \leq n$,

$$(AB)_{ij} = \sum_{r=1}^{m} (A)_{ir}(B)_{rj} = \sum_{r=1}^{m} 0 = 0.$$

All entries in row i of AB are zero. **The zero matrix** is a matrix of any specified shape whose entries are all zero.

🏆 **Victory!**

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5.4 Diagonal Matrices

A **diagonal matrix** is a **square** matrix whose off-diagonal entries are all zero. Write

$$D = \operatorname{diag}(d_1, \ldots, d_n) = \begin{pmatrix} d_1 & & 0 \\ & \ddots & \\ 0 & & d_n \end{pmatrix}.$$

Equivalently,

$$[D]_{ij} = \begin{cases} 0, & i \neq j, \\ d_i, & i = j. \end{cases}$$

A **scalar matrix** is a diagonal matrix with equal diagonal entries. The **identity matrix** is a diagonal matrix with all diagonal entries equal to 1, hence is a scalar matrix. A scalar matrix can therefore be written aI, where a is its common diagonal entry. Write I_n for the $n \times n$ identity, or simply I when its size is clear. For every $n \times m$ matrix A, $AI_m = I_n A = A$.

EXAMPLE 5.4

$$I_2 A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}, \qquad AI_2 = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}.$$

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EXERCISE 5.8

Let A, B be diagonal matrices of order n. Please prove that:

1. $A + B$ is diagonal.
2. AB is diagonal and $[AB]_{ii} = a_{ii}b_{ii}$ for every $1 \leq i \leq n$.
3. A and B **commute**: $AB = BA$.

WORKED SOLUTION

1. For $i \neq j$, $(A + B)_{ij} = a_{ij} + b_{ij} = 0 + 0 = 0$. Thus $A + B$ is diagonal, with diagonal entries $(A + B)_{ii} = a_{ii} + b_{ii}$.
2. By the definition of multiplication,

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj}.$$

   In row i of A, only a_{ii} can be nonzero, so the sum reduces to $a_{ii}b_{ij}$. If $i \neq j$, then $b_{ij} = 0$, giving $(AB)_{ij} = 0$. If $i = j$, then $(AB)_{ii} = a_{ii}b_{ii}$. Hence AB is diagonal.
3. Similarly BA is diagonal with diagonal entries $b_{ii}a_{ii}$. Scalars in a field commute, so

$$a_{ii}b_{ii} = b_{ii}a_{ii} \qquad (1 \leq i \leq n).$$

   All entries of AB and BA agree; therefore $AB = BA$.

And we're done. Victory!

NOTE

Check dimensions before computing. The product $A_{k\times m}B_{m\times n}$ has k rows and n columns. Equality of matrices requires both equal dimensions and equal corresponding entries.

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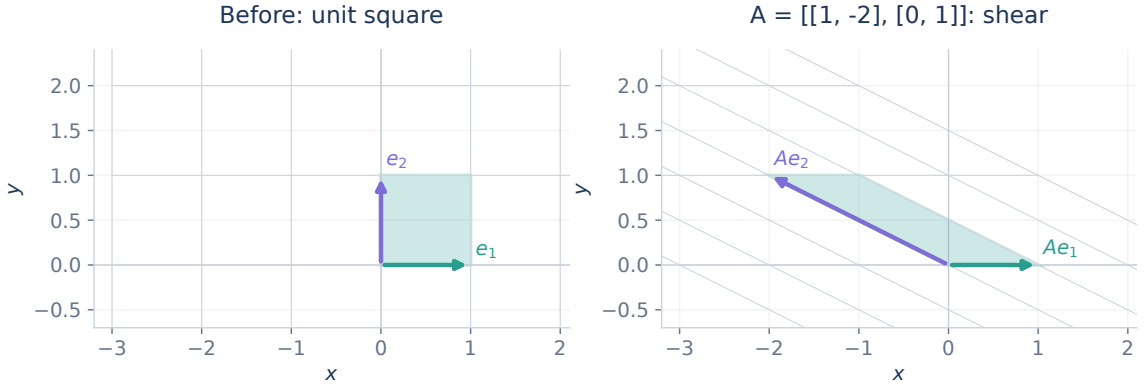


Figure 5.2. A linear map takes a straight grid to a straight grid. The columns of the matrix describe the images of the basis vectors.

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5.5 Transpose

- The transpose is defined for every matrix. The **transpose** A^T satisfies $(A^T)_{ij} = (A)_{ji}$.
- Rows become columns and columns become rows.
- If $A \in \mathbb{R}^{m\times n}$, then $A^T \in \mathbb{R}^{n\times m}$: the numbers of rows and columns are interchanged.

EXAMPLE 5.5

$$C = \begin{pmatrix} 2 & -3 \\ 0 & 7 \\ -4 & 1 \end{pmatrix} \in \mathbb{R}^{3\times 2}, \qquad C^T = \begin{pmatrix} 2 & 0 & -4 \\ -3 & 7 & 1 \end{pmatrix} \in \mathbb{R}^{2\times 3}.$$

The rows of C are the columns of C^T.

5.6 Symmetry and Skew-Symmetry

- A **square** matrix A is **symmetric** if $A^T = A$.
- A **square** matrix A is **skew-symmetric** if $A^T = -A$.
- Every diagonal matrix is symmetric; in particular, every scalar matrix and the identity matrix are symmetric.

The square condition matters: $A \in \mathbb{F}^{m\times n}$ and $A^T \in \mathbb{F}^{n\times m}$ have the same shape only when $n = m$. Intuitively, symmetry means reflection across the main diagonal; skew-symmetry means the same reflection followed by a change of sign.

EXERCISE 5.9

Please classify each matrix as symmetric, skew-symmetric, or neither.

$$A = \begin{pmatrix} 2 & -4 \\ -4 & 7 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 2 & 1 \\ -5 & 0 \end{pmatrix}.$$

WORKED SOLUTION

Computing transposes gives

$$A^T = \begin{pmatrix} 2 & -4 \\ -4 & 7 \end{pmatrix} = A, \qquad B^T = \begin{pmatrix} 0 & -3 \\ 3 & 0 \end{pmatrix} = -B,$$

$$C^T = \begin{pmatrix} 2 & -5 \\ 1 & 0 \end{pmatrix} \neq C, \qquad C^T \neq -C.$$

Thus A is symmetric, B is skew-symmetric, and C is neither.

You know what that means? V-I-C-T-O-R-Y!

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EXERCISE 5.10

Please prove or disprove.

1. If A, B are symmetric, then AB is symmetric.
2. If A, B are both nonsymmetric, then AB is nonsymmetric.

WORKED SOLUTION

1. $(AB)^T = B^T A^T = BA$, so symmetry of both factors does not ensure symmetry of the product. For example,

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & -1 \\ -1 & 3 \end{pmatrix}, \quad AB = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}.$$

   Both factors are symmetric, but the product is not.

2. Take $A = B = \begin{pmatrix} 0 & 2 \\ -3 & 0 \end{pmatrix}$. Neither is symmetric, but

$$AB = \begin{pmatrix} 0 \cdot 0 + 2(-3) & 0 \cdot 2 + 2 \cdot 0 \\ (-3)0 + 0(-3) & (-3)2 + 0 \cdot 0 \end{pmatrix} = -6I_2.$$

   The product is symmetric. Thus both assertions are false.

Another one down. Victory!

EXERCISE 5.11

Let $A \in M_n(\mathbb{F})$, where $\text{char}(\mathbb{F}) \neq 2$. Please prove that:

1. $A + A^T$ is symmetric.
2. $A - A^T$ is skew-symmetric.
3. A can be written as the sum of a symmetric matrix and a skew-symmetric matrix.

WORKED SOLUTION

1. $(A + A^T)^T = A^T + (A^T)^T = A^T + A = A + A^T$.
2. $(A - A^T)^T = A^T - A = -(A - A^T)$, proving skew-symmetry.
3. Add the matrices from the preceding parts:

$$2A = (A + A^T) + (A - A^T),$$
$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T).$$

   As shown above, the first summand is symmetric and the second is skew-symmetric.

We've got it. Victory!

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EXERCISE 5.12

$$D = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \qquad H = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}.$$

Please compute DH and HD entry by entry, please compare their action on e_2, and please find all v satisfying $DHv = HDv$.

WORKED SOLUTION

$$DH = \begin{pmatrix} 2\cdot 1 + 0\cdot 0 & 2(-1) + 0\cdot 1 \\ 0\cdot 1 + 1\cdot 0 & 0(-1) + 1\cdot 1 \end{pmatrix} = \begin{pmatrix} 2 & -2 \\ 0 & 1 \end{pmatrix},$$

$$HD = \begin{pmatrix} 1\cdot 2 + (-1)0 & 1\cdot 0 + (-1)1 \\ 0\cdot 2 + 1\cdot 0 & 0\cdot 0 + 1\cdot 1 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 0 & 1 \end{pmatrix}.$$

Thus $DHe_2 = (-2, 1)^T$, whereas $HDe_2 = (-1, 1)^T$. For $v = (x, y)^T$,

$$DHv - HDv = (2x - 2y - (2x - y),\ y - y)^T = (-y, 0)^T.$$

The results agree exactly when $y = 0$, on $\mathrm{span}\{e_1\}$.

That does it. Victory!

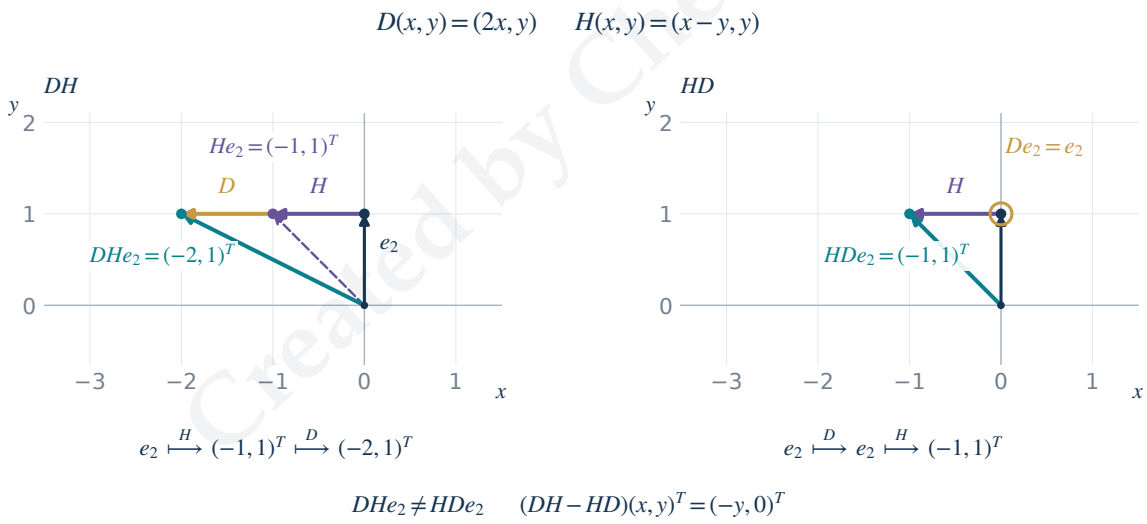


Figure 5.3. Operation order on the same vector: DH applies H first, whereas HD applies D first. The ring marks the fixed point $De_2 = e_2$, but the final results differ: $DHe_2 = (-2, 1)^T \neq (-1, 1)^T = HDe_2$.

Additional practice

EXERCISE 5.13 FOUNDATION

$$A = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 1 \\ -1 & 0 \\ 0 & 3 \end{pmatrix}.$$

Please determine the dimensions of AB, BA and please compute both. Please use $e_2 \in \mathbb{R}^2$ to verify $(AB)e_2 = A(Be_2)$. Can $AB = BA$?

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EXERCISE 5.13 FOUNDATION CONTINUED

WORKED SOLUTION

The dimensions are 2×2 and 3×3, respectively.

$$AB = \begin{pmatrix} 1\cdot 2 + 0(-1) - 1\cdot 0 & 1\cdot 1 + 0\cdot 0 - 1\cdot 3 \\ 0\cdot 2 + 2(-1) + 1\cdot 0 & 0\cdot 1 + 2\cdot 0 + 1\cdot 3 \end{pmatrix} = \begin{pmatrix} 2 & -2 \\ -2 & 3 \end{pmatrix},$$

$$BA = \begin{pmatrix} 2\cdot 1 + 1\cdot 0 & 2\cdot 0 + 1\cdot 2 & 2(-1) + 1\cdot 1 \\ (-1)1 + 0\cdot 0 & (-1)0 + 0\cdot 2 & (-1)(-1) + 0\cdot 1 \\ 0\cdot 1 + 3\cdot 0 & 0\cdot 0 + 3\cdot 2 & 0(-1) + 3\cdot 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 & -1 \\ -1 & 0 & 1 \\ 0 & 6 & 3 \end{pmatrix}.$$

Here $Be_2 = (1, 0, 3)^T$, so $A(Be_2) = (-2, 3)^T$. Also $(AB)e_2 = (-2, 3)^T$, the second column of AB. The dimensions differ, so AB and BA cannot be equal.

All wrapped up. Victory!

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EXERCISE 5.14 CHALLENGE

Let $n \geq 2$ and $A \in M_n(\mathbb{F})$ over an arbitrary field. Please prove:

1. A commutes with every diagonal matrix if and only if A is diagonal.
2. A commutes with every matrix in $M_n(\mathbb{F})$ if and only if $A = \lambda I_n$ for some $\lambda \in \mathbb{F}$.

You may use the matrix units E_{ij}, whose entry (i, j) is 1 and all other entries are zero.

WORKED SOLUTION

Suppose A commutes with every diagonal matrix. In particular, $AE_{kk} = E_{kk}A$ for every k. In entry (i, k) with $i \neq k$, the left side has value a_{ik} and the right side has value zero. Hence $a_{ik} = 0$ for every $i \neq k$, so A is diagonal. Conversely, two diagonal matrices commute because their product is diagonal and its diagonal entries are products of commuting scalars.
If A commutes with every matrix, part 1 gives $A = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. For $i \neq j$,

$$AE_{ij} = \lambda_i E_{ij}, \qquad E_{ij}A = \lambda_j E_{ij}.$$

Comparing entry (i, j) gives $\lambda_i = \lambda_j$. All diagonal entries are therefore equal, so $A = \lambda I_n$. Conversely, for every B, $(\lambda I_n)B = \lambda B = B(\lambda I_n)$. The proof uses only the scalars $0, 1$ and field multiplication, so it applies to finite fields as well.

We've reached the finish. Victory!

NOTATION AND CONVENTIONS

For $1 \leq i \leq m$ and $1 \leq j \leq n$, the matrix $E_{ij} \in M_{m\times n}(F)$ has entry 1 at (i, j) and zero elsewhere. Every matrix $A = (a_{ij})$ has a unique representation $A = \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij} E_{ij}$. Matrix addition and scalar multiplication therefore act componentwise, just as for columns in F^{mn}.

EXERCISE 5.15

Please determine whether A lies in the span of B_1, B_2 over $\mathbb{R}$.

$$A = \begin{pmatrix} -2 & 4 \\ 3 & -3 \end{pmatrix}, \qquad B_1 = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}, \qquad B_2 = \begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix}.$$

WORKED SOLUTION

$$aB_1 + bB_2 = \begin{pmatrix} a & -a \\ b & -b \end{pmatrix}.$$

Equality with A would require $a = -2$ and $-a = 4$, hence $a = -4$. This contradiction proves that A is not in the span. The choice $b = 3$ matches only the second row.

Yes! Victory!

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EXERCISE 5.16

Let $A = \begin{pmatrix} 1 & -2 \\ 0 & 0 \end{pmatrix}$ over $\mathbb{R}$. Please characterize all matrices B satisfying $AB = BA$, and please show that they are precisely the linear combinations of A and $I_2 - A$.

WORKED SOLUTION

Write $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Computing both products gives

$$AB = \begin{pmatrix} a - 2c & b - 2d \\ 0 & 0 \end{pmatrix}, \qquad BA = \begin{pmatrix} a & -2a \\ c & -2c \end{pmatrix}.$$

Equality requires $c = 0$ and $b = -2(a - d)$. Hence

$$B = \begin{pmatrix} a & -2(a-d) \\ 0 & d \end{pmatrix} = aA + d(I_2 - A).$$

Substitution into both products shows conversely that every matrix of this form commutes with A.

V-I-C-T-O-R-Y!

EXERCISE 5.17

Work in $M_3(\mathbb{R})$. The matrix unit E_{ij} has entry 1 at (i, j) and zero elsewhere. Define

$$D_i = E_{ii} \quad (1 \le i \le 3), \qquad S_{ij} = E_{ij} + E_{ji}, \quad K_{ij} = E_{ij} - E_{ji} \quad (1 \le i < j \le 3).$$

$$\mathcal{B}_s = \{D_1, D_2, D_3, S_{12}, S_{13}, S_{23}\}, \qquad \mathcal{B}_a = \{K_{12}, K_{13}, K_{23}\}.$$

Please prove that $\mathcal{B}_s$ is a basis of the symmetric matrices and $\mathcal{B}_a$ is a basis of the skew-symmetric matrices. Please prove that their union is a basis of $M_3(\mathbb{R})$ and please compute the three dimensions.

WORKED SOLUTION

We have $D_i^T = D_i$, $S_{ij}^T = S_{ij}$, and $K_{ij}^T = -K_{ij}$. Every symmetric $A = (a_{ij})$ has the representation

$$A = \sum_{i=1}^{3} a_{ii} D_i + \sum_{1 \le i < j \le 3} a_{ij} S_{ij}.$$

The diagonal and upper-diagonal entries determine all coefficients uniquely. Thus the family spans and is independent, giving a basis of size $3 + 3 = 6$.
For skew-symmetric A, the diagonal is zero and $a_{ji} = -a_{ij}$. Hence

$$A = \sum_{1 \le i < j \le 3} a_{ij} K_{ij}.$$

Again, the entries above the diagonal uniquely determine the coefficients, so $\mathcal{B}_a$ is a basis of size 3.

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EXERCISE 5.17 CONTINUED

Every $A \in M_3(\mathbb{R})$ satisfies

$$A = \frac{A + A^T}{2} + \frac{A - A^T}{2}.$$

The first summand is symmetric and the second skew-symmetric, so the union spans all matrices. The intersection of the two subspaces is $\{0\}$: $A^T = A = -A$ gives $2A = 0$, hence $A = 0$. Any zero combination of the union therefore splits into a zero combination in each basis, forcing every coefficient to vanish. The union is a basis, with dimensions

$$\dim \mathrm{Sym}_3(\mathbb{R}) = 6, \quad \dim \mathrm{ASym}_3(\mathbb{R}) = 3, \quad \dim M_3(\mathbb{R}) = 9 = 6 + 3.$$

Victory!

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EXERCISE 5.18 CHALLENGE

Let $a_1, \ldots, a_{n+2} \in \mathbb{R}^n$, where $n \geq 1$. For a nonempty index set I, define the convex hull by

$$\operatorname{conv}\{a_i : i \in I\} = \left\{ \sum_{i \in I} \lambda_i a_i : \lambda_i \geq 0,\ \sum_{i \in I} \lambda_i = 1 \right\}.$$

Please prove that the indices can be partitioned into two nonempty sets I, J whose convex hulls intersect.

WORKED SOLUTION

Consider real coefficients $\alpha_1, \ldots, \alpha_{n+2}$ satisfying

$$\sum_{i=1}^{n+2} \alpha_i a_i = 0, \qquad \sum_{i=1}^{n+2} \alpha_i = 0.$$

This is a homogeneous system of $n+1$ equations in $n+2$ unknowns, so it has a nonzero solution. Since the coefficients sum to zero, at least one is positive and at least one is negative. Put

$$I = \{i : \alpha_i \geq 0\}, \qquad J = \{i : \alpha_i < 0\}, \qquad s = \sum_{i \in I} \alpha_i = -\sum_{j \in J} \alpha_j > 0.$$

These sets are nonempty and partition all indices. The vector equation yields

$$\sum_{i \in I} \frac{\alpha_i}{s} a_i = \sum_{j \in J} \frac{-\alpha_j}{s} a_j.$$

On each side the coefficients are nonnegative and sum to 1. The common point therefore lies in both convex hulls.

And we're done. Victory!

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